Programming Fundamentals

Lab 5

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Program: BS Computer Science (BSCS)

Section: D

University: Ghazi University

Task 1:

C++ program to Print Table of any Number.

The concept of generating table of any number is multiply particular number from 1 to 10.

Input

Task_1.cpp

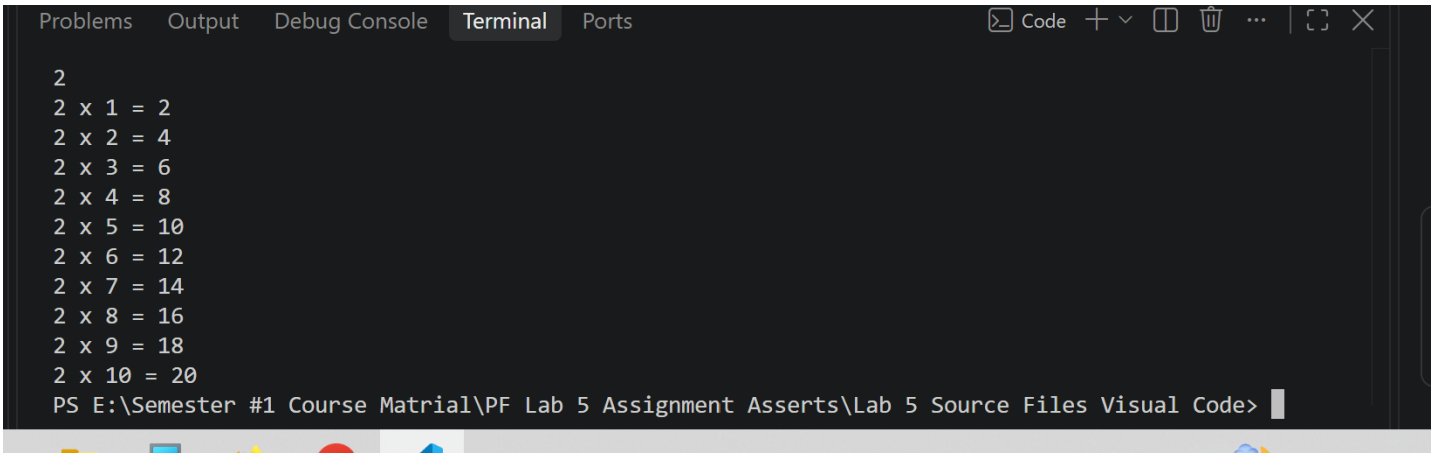
```cpp
#include <iostream>
using namespace std;

int main()
{
    int n;
    cin >> n;

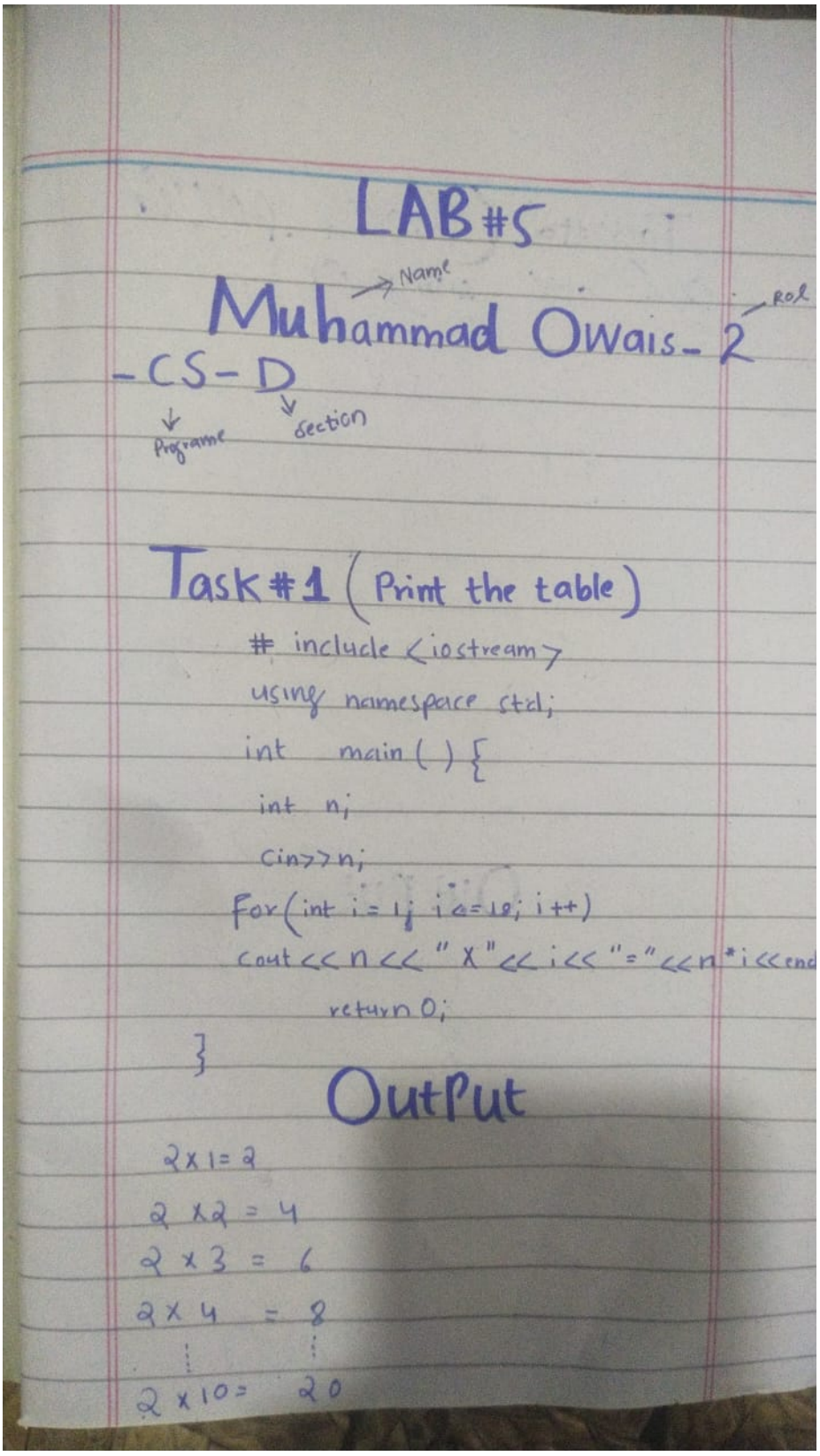
    for (int i = 1; i <= 10; i++)
        cout << n << " x " << i << " = " << n * i << endl;

    return 0;
}
```

Output



Handwritten Notes



Task 2:

Find power of any number using for loop.

Input

Task_1.cpp Task_2.cpp

Task_2.cpp > main()

```cpp
#include <iostream>
using namespace std;

int main()
{
    int a, b, p = 1;
    cin >> a >> b;

    for (int i = 1; i <= b; i++)
        p *= a;

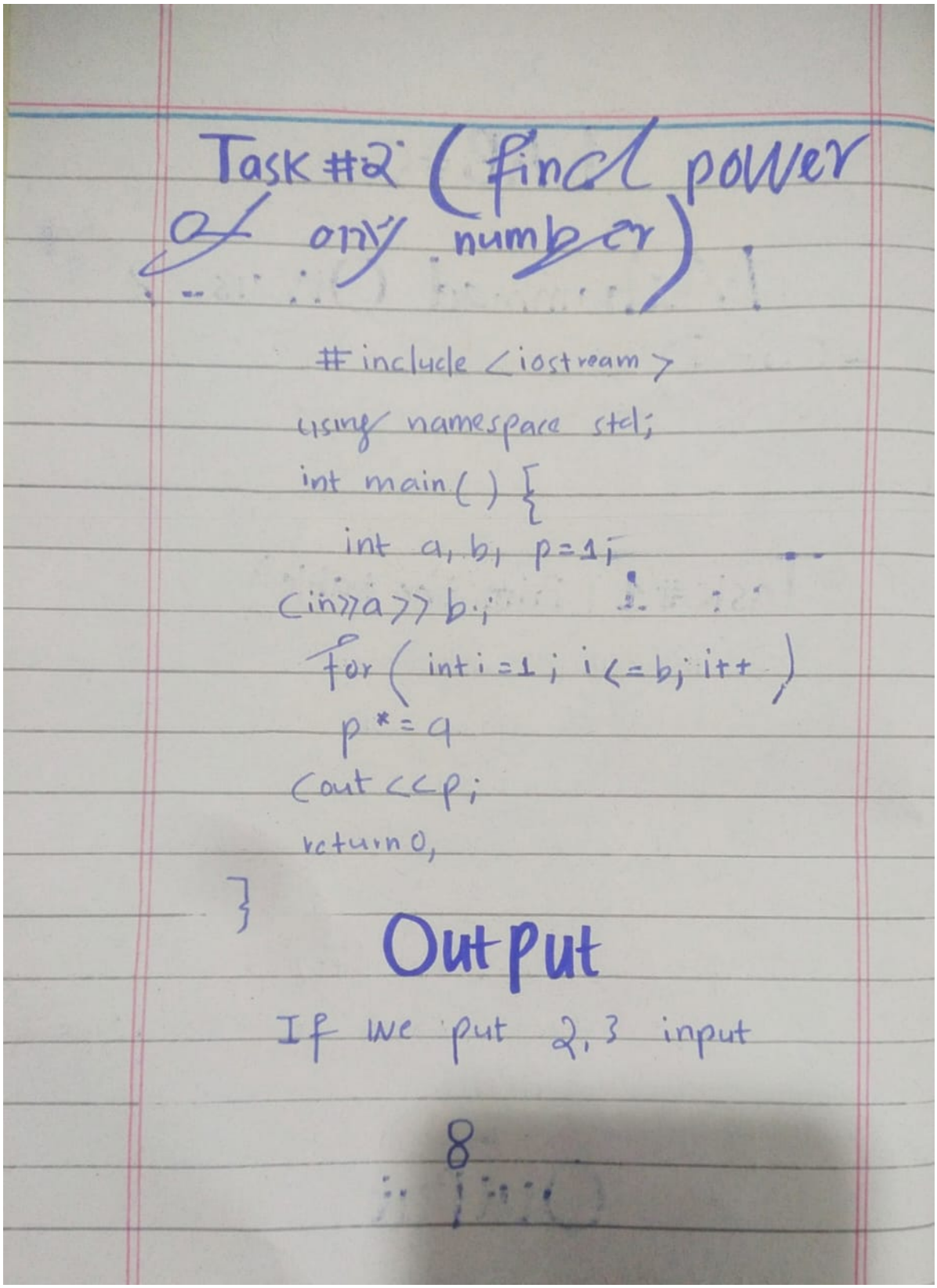
    cout << p;
    return 0;
}
```

Output

Problems Output Debug Console Terminal Ports Code

```
PS E:\Semester #1 Course Matrial\PF Lab 5 Assignment Asserts\Lab 5 Source Files Visual Code> cd "e:\
Semester #1 Course Matrial\PF Lab 5 Assignment Asserts\Lab 5 Source Files Visual Code\" ; if ($?) {
g++ Task_2.cpp -o Task_2 } ; if ($?) { .\Task_2 }
2 3
8
PS E:\Semester #1 Course Matrial\PF Lab 5 Assignment Asserts\Lab 5 Source Files Visual Code>
```

Handwritten Notes



Task #2 (final power of any number)

```
#include <iostream>
using namespace std;
int main() {
  int a, b, p=1;
cin>>a>>b;
 for( int i=1; i<=b; i++ )
  p*=a
cout<<p;
return 0,
}
```

Output

If we put 2, 3 input

8

Task 3:

Sum of Natural Numbers using loop.

Input

Task_1.cpp | Task_2.cpp | Task_3.cpp

Task_3.cpp > main()

```cpp
#include <iostream>
using namespace std;

int main()
{
    int n, sum = 0;
    cin >> n;

    for (int i = 1; i <= n; i++)
        sum += i;

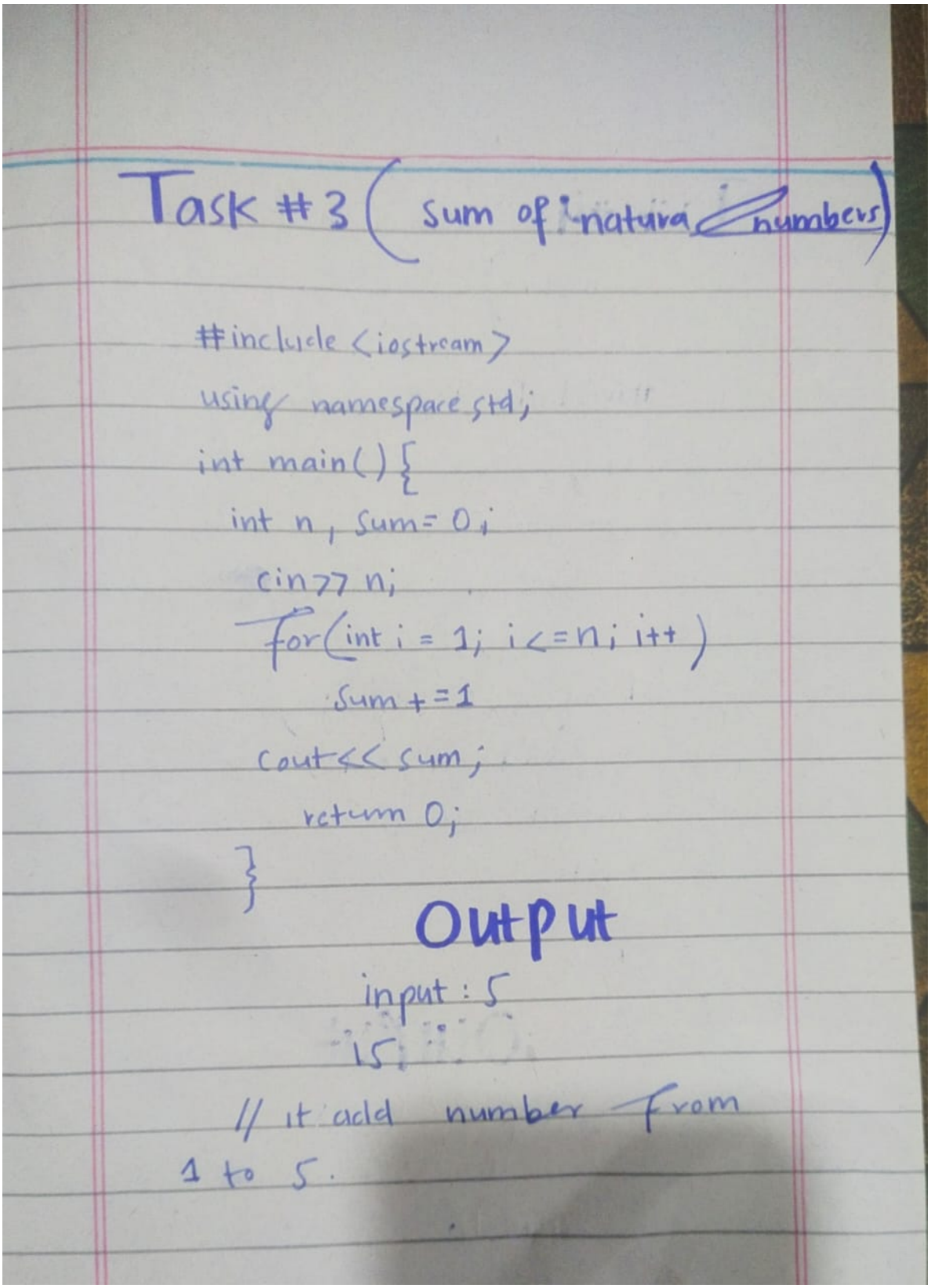
    cout << sum;
    return 0;
}
```

Output

Problems | Output | Debug Console | Terminal | Ports

```
PS E:\Semester #1 Course Matrial\PF Lab 5 Assignment Asserts\Lab 5 Source Files Visual Code> cd "e:\
Semester #1 Course Matrial\PF Lab 5 Assignment Asserts\Lab 5 Source Files Visual Code\" ; if ($?) {
g++ Task_3.cpp -o Task_3 } ; if ($?) { .\Task_3 }
5
15
PS E:\Semester #1 Course Matrial\PF Lab 5 Assignment Asserts\Lab 5 Source Files Visual Code>
```

Handwritten Notes



Task #3 (sum of natural numbers)

```
#include <iostream>
using namespace std;
int main() {
  int n, sum = 0;
  cin >> n;
  for(int i = 1; i <= n; i++)
      sum += 1
  cout << sum;
    return 0;
}
```

Output

input : 5

15

// it add number from 1 to 5.

Task 4:

C++ program to find sum of digits of a number.

Sum of digits means add all the digits of any number, for example we take any number like 358. Its sum of all digit is 3+5+8=16. Using given code we can easily write c++ program.

Input

Task_1.cpp | Task_2.cpp | Task_3.cpp | Task_4.cpp

Task_4.cpp > main()

```cpp
#include <iostream>
using namespace std;

int main()
{
    int n, sum = 0;
    cin >> n;

    for (; n > 0; n /= 10)
        sum += n % 10;

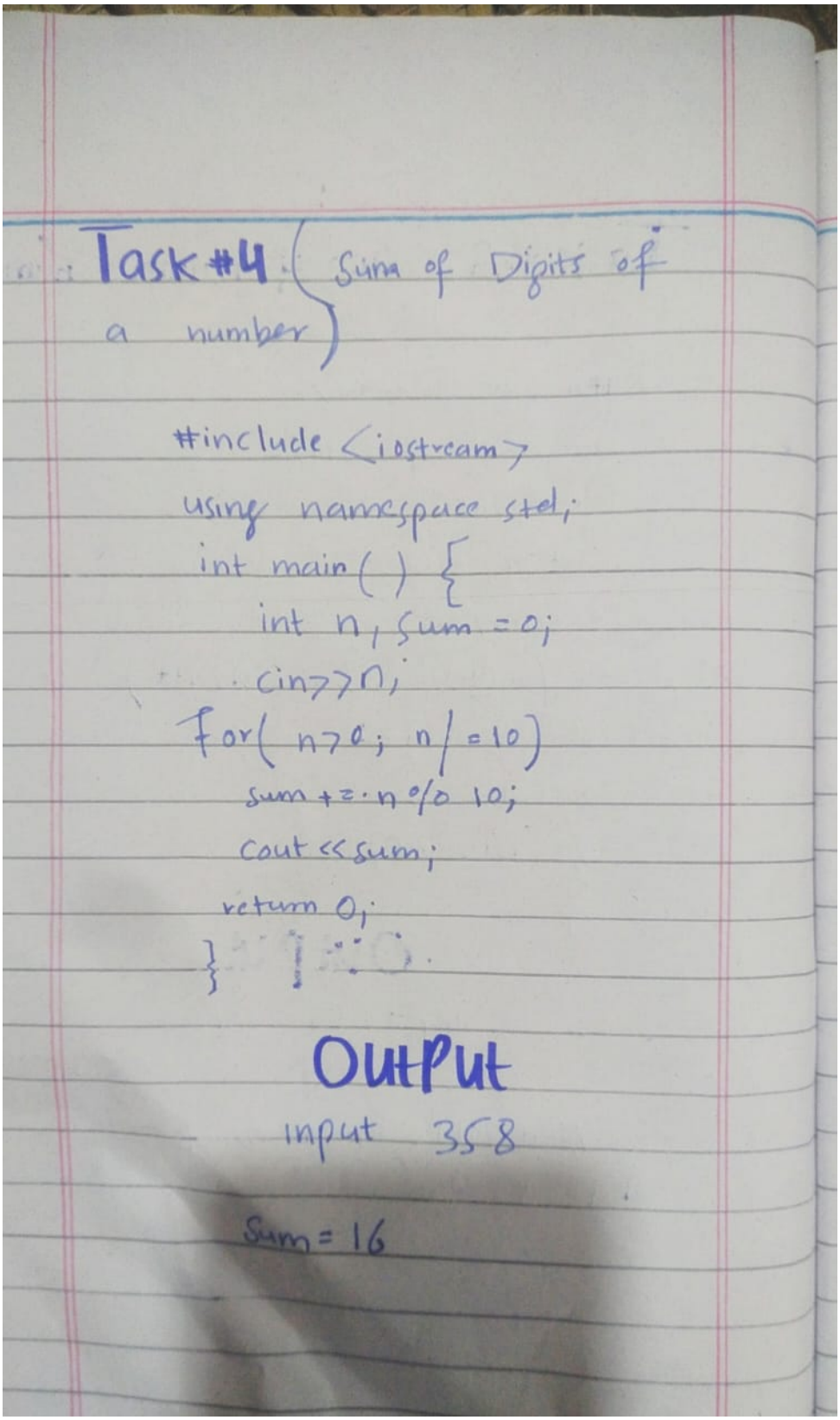
    cout << sum;
    return 0;
}
```

Output

Problems | Output | Debug Console | Terminal | Ports

```
PS E:\Semester #1 Course Matrial\PF Lab 5 Assignment Asserts\Lab 5 Source Files Visual Code> cd "e:\
Semester #1 Course Matrial\PF Lab 5 Assignment Asserts\Lab 5 Source Files Visual Code\" ; if ($?) {
g++ Task_4.cpp -o Task_4 } ; if ($?) { .\Task_4 }
358
16
PS E:\Semester #1 Course Matrial\PF Lab 5 Assignment Asserts\Lab 5 Source Files Visual Code>
```

Handwritten Notes



Task #4 (Sum of Digits of a number)

```
#include <iostream>
using namespace std;
int main( ) {
    int n, sum = 0;
    cin>>n;
for( n>0; n/=10)
    sum += n%10;
    cout<<sum;
    return 0;
}
```

Output

input 358

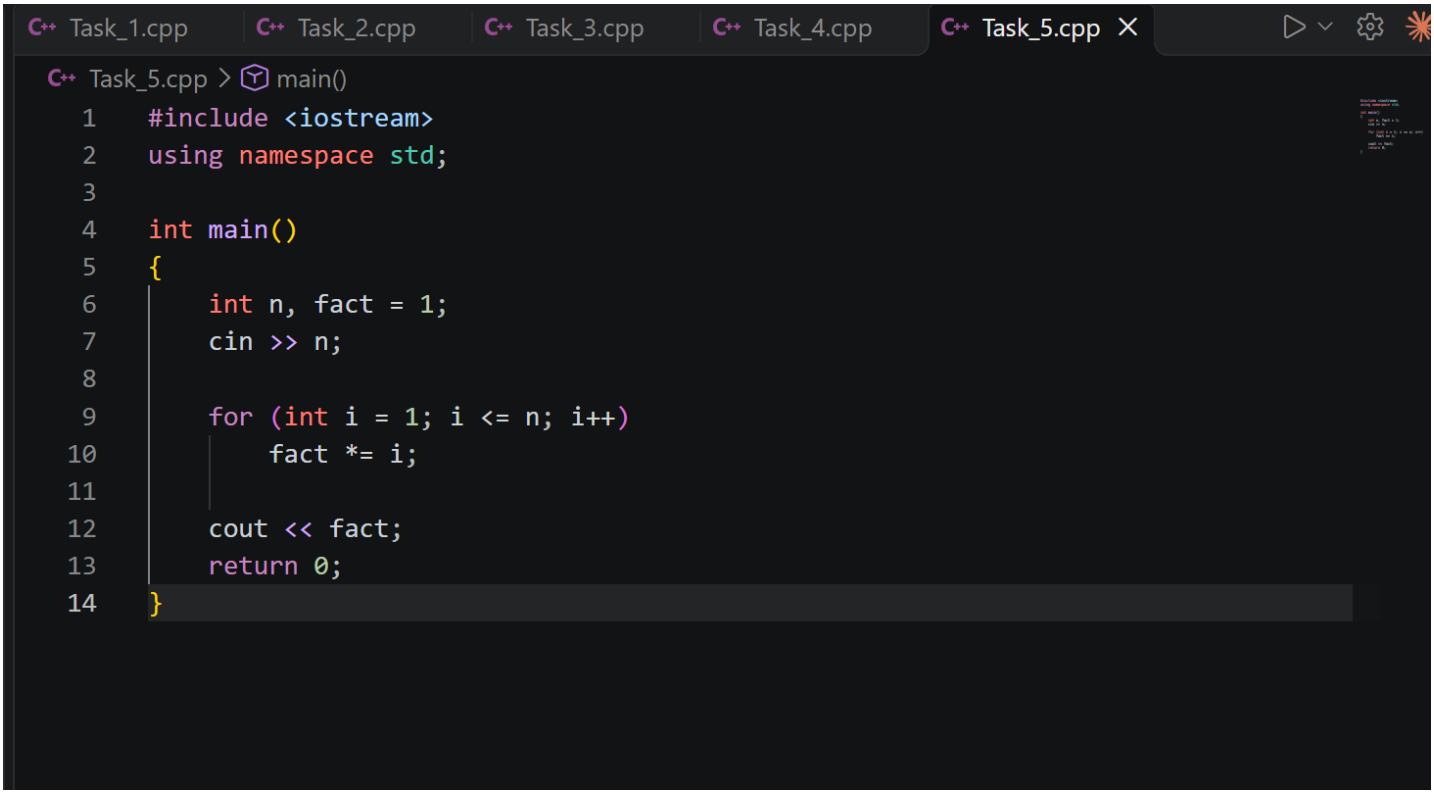
Sum = 16

Task 5:

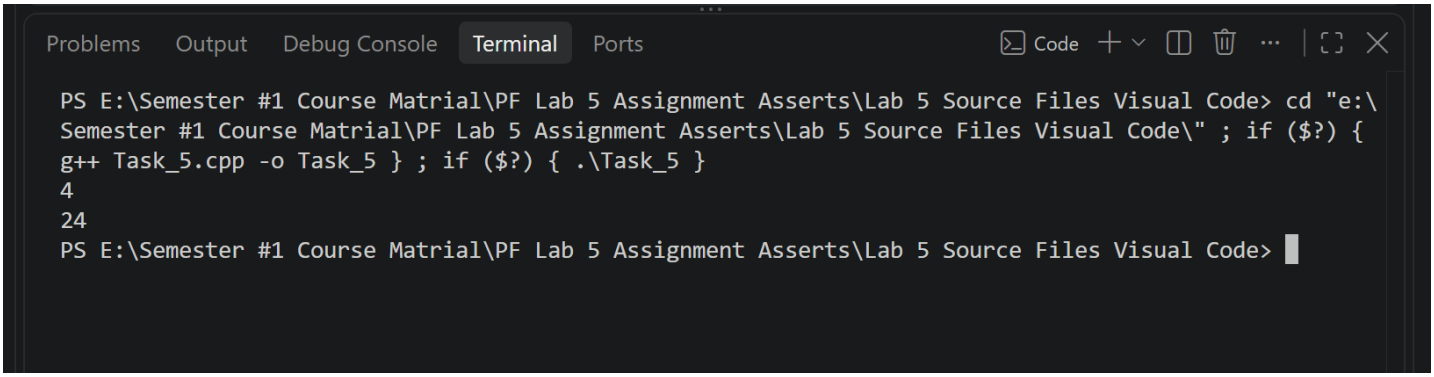
Program to Generate Factorial a Certain Number

Factorial of any number is the product of an integer and all the integers below it for example factorial of 4 is 4! = 4 * 3 * 2 * 1 = 24

Input



Output



Handwritten Notes

This handwritten page is shared with the other task below: it contains both Task #5 and Task #7.

Task#5 (factorial of number)

```
#include<iostream>
using namespace std;
int main(){
  int n, fact=1;
  cin>>n;
for( int i=1; i<=n; i++ )
  fact*=1;
  return 0;
}
```

Output

input: 4

fact= 24

Task #7 (print star pattern)

```
#include<iostream>
using namespace std;
 int main( ) {
for( int i=1; i<=7; i++)
  {
  for( int j=1; j<=i; j++)
  cout<<endl;
    return 0;
}
```

Output

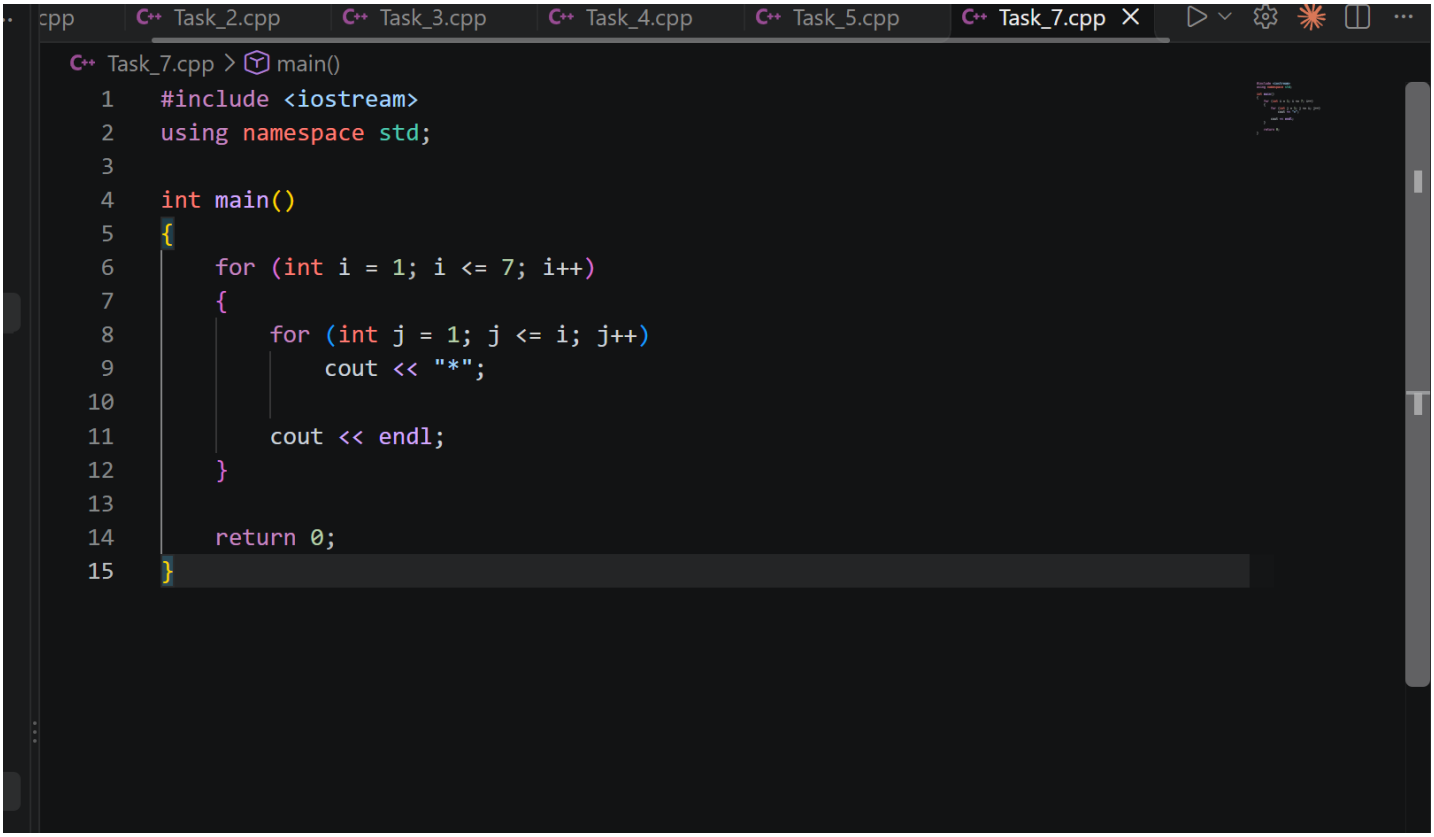
```
*
**
***
:
*******
```

Task 7:

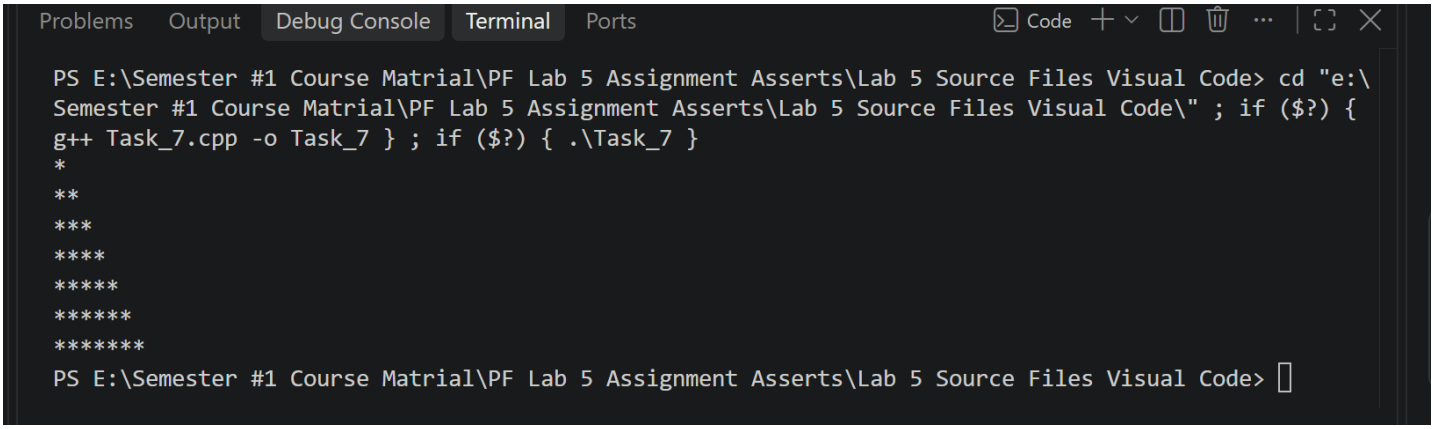
Write a C++ program that will print the following pattern.

```
*
**
***
****
*****
******
*******
```

Input



Output



Handwritten Notes

This handwritten page is shared with the other task below: it contains both Task #5 and Task #7.

Task#5 (factorial of number)

```
#include<iostream>
using namespace std;
int main(){
  int n, fact=1;
  cin>>n;
for( int i=1; i<=n; i++ )
  fact*=1;
  return 0;
}
```

Output

input: 4

fact= 24

Task #7 (print star pattern)

```
#include<iostream>
using namespace std;
 int main( ) {
for( int i=1; i<=7; i++)
  {
  for( int j=1; j<=i; j++)
  cout<<endl;
    return 0;
}
```

Output

```
*
**
***
:
*******
```

Task 8:

Write a C program print total number of days in a month using switch case.

Input

Task_8.cpp > main()

```cpp
#include <iostream>
using namespace std;

int main()
{
    int m;
    cin >> m;

    switch (m)
    {
        case 2: cout << 28; break;
        case 4: case 6: case 9: case 11:
            cout << 30; break;
        case 1: case 3: case 5: case 7:
        case 8: case 10: case 12:
            cout << 31; break;
        default: cout << "Invalid month";
    }

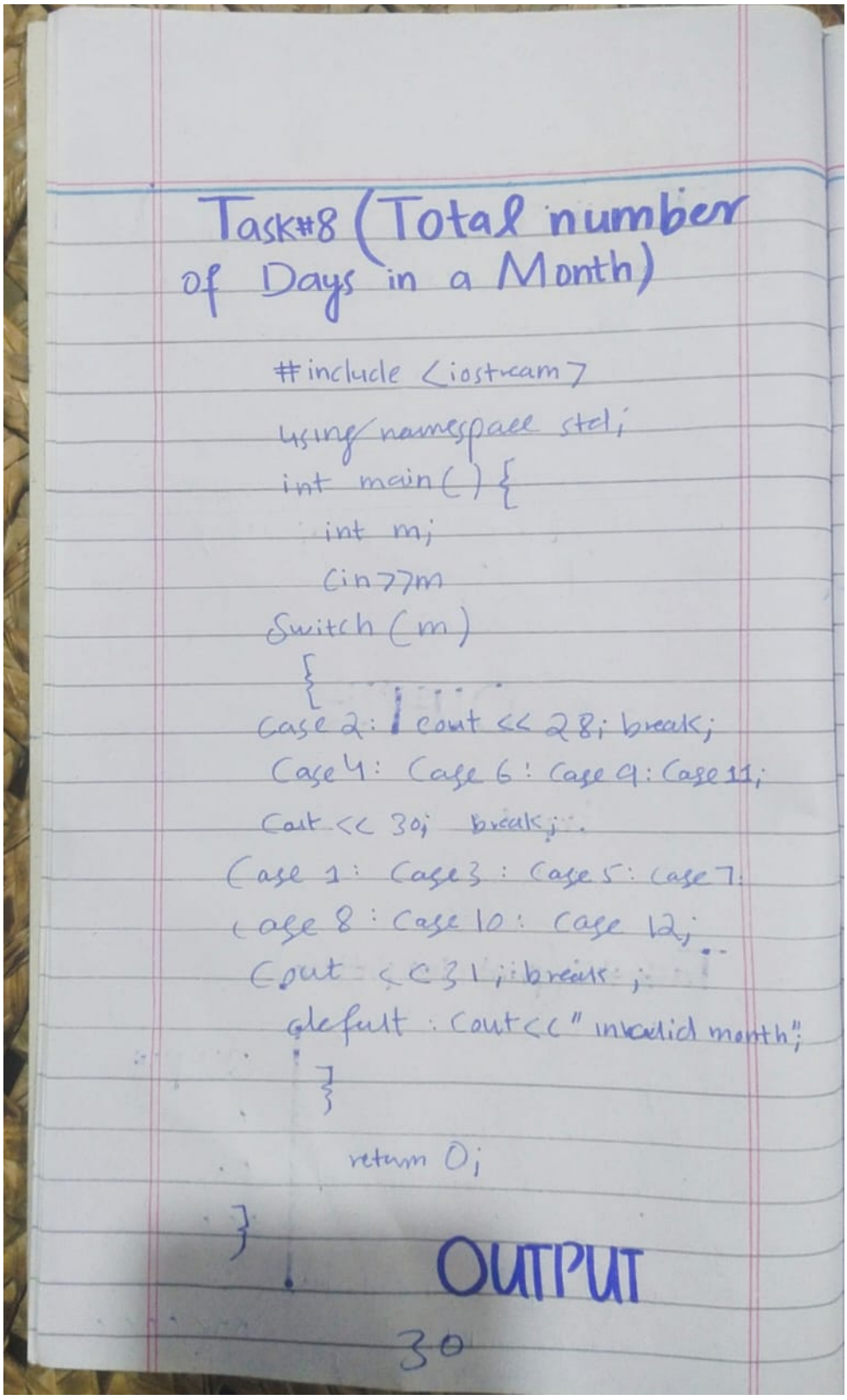
    return 0;
}
```

Output

Problems Output Debug Console Terminal Ports Code

```
PS E:\Semester #1 Course Matrial\PF Lab 5 Assignment Asserts\Lab 5 Source Files Visual Code> cd "e:\
Semester #1 Course Matrial\PF Lab 5 Assignment Asserts\Lab 5 Source Files Visual Code\" ; if ($?) {
g++ Task_8.cpp -o Task_8 } ; if ($?) { .\Task_8 }
4
30
PS E:\Semester #1 Course Matrial\PF Lab 5 Assignment Asserts\Lab 5 Source Files Visual Code>
```

Handwritten Notes



Task#8 (Total number of Days in a Month)

```
#include <iostream>
using namespace std;
int main(){
  int m;
   cin>>m
Switch(m)
  {
case 2: cout << 28; break;
 Case 4: Case 6: Case 9: Case 11;
Cout << 30; break;
Case 1: Case 3: Case 5: Case 7:
Case 8: Case 10: Case 12;
Cout << 31; break;
 default : Cout<<" invalid month";
  }
    return 0;
}
```

OUTPUT

30

Task 9:

Write a C program to check whether an alphabet is vowel or consonant using switch case.

Input

```cpp
#include <iostream>
using namespace std;

int main()
{
    char ch;
    cin >> ch;

    switch (ch)
    {
        case 'a': case 'e': case 'i':
        case 'o': case 'u':
        case 'A': case 'E': case 'I':
        case 'O': case 'U':
            cout << "Vowel";
            break;
        default:
            cout << "Consonant";
    }

    return 0;
}
```

Output

```
PS E:\Semester #1 Course Matrial\PF Lab 5 Assignment Asserts\Lab 5 Source Files Visual Code> cd "e:\Semester #1 Course Matrial\PF Lab 5 Assignment Asserts\Lab 5 Source Files Visual Code\" ; if ($?) { g++ Task_9.cpp -o Task_9 } ; if ($?) { .\Task_9 }
a
Vowel
PS E:\Semester #1 Course Matrial\PF Lab 5 Assignment Asserts\Lab 5 Source Files Visual Code>
```

Handwritten Notes

